

Advanced Algebra & Systems



Systems of three linear equations

- 1 Solve the system: $x + y + z = 6$, $2x - y + z = 3$, $x + 2y - z = 5$.
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Solving polynomial equations of higher degree

- 2 Solve $x^3 - 6x^2 + 11x - 6 = 0$, given that $x = 1$ is a root.
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Rational root theorem

- 3 List the possible rational roots of $2x^3 - 3x^2 - 8x + 12 = 0$, using the rational root theorem, then find one actual root by testing.
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Partial fractions

- 4 Express $\frac{5x - 1}{(x - 1)(x + 2)}$ in partial fractions.
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Radical equations with extraneous solutions

- 5 Solve $\sqrt{2x + 7} = x + 2$, and check for extraneous solutions.
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Systems with a quadratic and a linear equation

- 6 Solve the system: $y = x^2 - 3x - 4$ and $y = 2x - 4$.
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Logarithmic and exponential equations

- 7 Solve $\log_2(x + 3) = 5$.
- 8 Solve $3^{2x-1} = 27$.
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Inequalities involving polynomials

- 9 Solve $x^2 - x - 6 > 0$, giving your answer using interval notation reasoning.
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A multi-part system and polynomial combination problem

- 10 This question has two parts. A cubic polynomial $f(x) = x^3 + ax^2 + bx - 6$ has $x = 1$ and $x = 2$ as roots. (a) Use the factor theorem to set up and solve a system for a and b . (b) Find the third root of the polynomial.
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Synthetic division

- 11 Use synthetic division to divide $2x^3 - 5x^2 + 3x - 1$ by $(x - 2)$, and state the quotient and remainder.
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Nonlinear systems with three variables

- 12 Solve the system: $x^2 + y^2 = 25$ and $y = x + 1$.
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Simplifying complex rational expressions

- 13 Simplify $\frac{\frac{1}{x} + \frac{1}{y}}{\frac{1}{x} - \frac{1}{y}}$.

- 14 Simplify $\frac{x^2 - 4}{x^2 - 5x + 6}$.
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Solving equations with rational exponents

- 15 Solve $x^{2/3} = 9$ for x .
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A multi-part rational function and system problem

- 16 This question has three parts. A boat's fuel efficiency is modeled by $E(v) = \frac{600}{v} + 0.02v^2$, where v is speed in mph and E is fuel used (gallons) for a fixed trip. (a) Find $E(20)$. (b) Find $E(30)$. (c) Based on parts (a) and (b), determine whether increasing speed from 20 to 30 mph increases or decreases fuel usage for this trip, and explain the two competing effects in the formula that cause this.
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Solving a system involving a rational equation

17 Solve the system: $\frac{1}{x} + \frac{1}{y} = \frac{5}{6}$ and $x + y = 5$.